

Data Augmentation for Imbalanced Regression: A Survey and Tutorial for Applied Econometrics

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Survey and Tutorial

Abstract

Imbalanced regression—where certain ranges of a continuous outcome or key covariates are under-represented in the sample—is pervasive in applied economics. Income distributions have sparse upper tails, catastrophic insurance claims are rare, and extreme poverty is both uncommon and policy-critical. Standard regression methods allocate estimation effort proportional to data density, performing poorly on these sparse but important regions. The machine learning community has developed a rich toolkit of *data augmentation* methods to address this problem, but these methods remain largely unknown to applied econometricians.

This paper provides a comprehensive survey and tutorial introducing these techniques to an economics audience. We catalogue augmentation methods—from synthetic oversampling (SMOTER, SMOGN) and statistical generators (KDE, copulas) to deep generative models (GANs, VAEs) and hybrid augmentation-weighting approaches—describing each with consistent notation and connecting them to familiar econometric concepts such as attenuation bias, sample selection, importance weighting, and bootstrap resampling. We provide formal conditions under which augmentation reduces bias, present a multi-scenario simulation study comparing all methods alongside non-augmentation alternatives (quantile regression, weighted least squares), and discuss applications across six economic domains. A practical decision framework guides practitioners in diagnosing imbalance, selecting methods, and validating results.

Our simulation results show that simpler augmentation methods (SMOGN, KDE) often suffice, that the hybrid augmentation-plus-weighting approach yields the largest

*Comments welcome: wernerhl@gmail.com. A companion paper [[Hernani-Limarino, 2026](#)] applies the methods surveyed here to poverty prediction and Proxy Means Testing.

bias reductions when a target distribution is available, and that deep generative models frequently underperform simpler alternatives on the sample sizes typical in economics. We discuss these findings with an emphasis on honest assessment of limitations. Applications to poverty prediction—developed in detail in a companion paper [[Hernani-Limarino, 2026](#)][—]insurance, health economics, environmental risk, finance, and labor economics illustrate the breadth of the toolkit’s relevance.

Keywords: imbalanced regression, data augmentation, synthetic oversampling, measurement error, attenuation bias, SMOTE, generative models, tail prediction, applied econometrics

JEL Classification: C14, C15, C21, C53, C55, I32

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1 Introduction

Applied economists routinely encounter regression problems where certain ranges of the outcome variable or key covariates are under-represented in the available data. Income and consumption distributions exhibit heavy right tails: most households cluster in the middle, while the extremes—both the very poor and the very wealthy—are sparse. Insurance claim severity data are dominated by small, routine claims, with catastrophic losses occurring rarely but mattering enormously for risk management. Health expenditure data follow a similar pattern: most individuals incur modest costs, while a small fraction generates enormous bills that drive aggregate spending. In environmental economics, the damages from pollution or natural disasters concentrate in rare extreme events. In financial risk management, default events are infrequent but consequential. In labor economics, extreme wages—whether at the top or bottom of the distribution—are both rare and policy-relevant.

In all these settings, the standard econometric toolkit is poorly suited to the task. Ordinary least squares minimizes overall mean squared error, allocating estimation effort roughly in proportion to the density of observations. Regions of the outcome space with many observations are fitted well; regions with few observations receive little attention. When the policy question concerns precisely those sparse regions—*What predicts extreme poverty? What drives catastrophic insurance losses? Which households face the highest health expenditure risk?*—the standard estimator optimizes the wrong objective. The consequences are predictable: attenuated coefficient estimates, poor predictions in the tails, and systematic under-performance on the cases that matter most.

The machine learning community has recognized this problem under the banner of *imbalanced learning*. In classification, where the issue manifests as class imbalance (e.g., rare fraud cases versus many normal transactions), a large literature has developed remedies ranging from the seminal SMOTE algorithm [Chawla et al., 2002] to modern generative approaches. For regression with continuous outcomes, analogous methods have emerged more recently, including SMOTER [Torgo et al., 2013], SMOGN [Branco et al., 2017], and various generative data augmentation strategies. These methods synthesize additional observations in the sparse regions of the data to rebalance the training distribution, thereby improving model performance where it matters most.

Despite their relevance, these tools remain largely unknown in economics. Applied econometricians facing imbalanced regression problems typically resort to ad hoc solutions—trimming outliers, applying log transformations, or switching to quantile regression—without access to the systematic toolkit that the ML literature provides. This survey aims to bridge that gap.

1.1 What This Paper Is and Is Not

This paper is a **survey and tutorial**. Its contributions are pedagogical and organizational:

- (i) **Translation:** We introduce ML concepts of imbalanced regression using econometric language and connect them to familiar problems (attenuation bias, sample selection, importance weighting).
- (ii) **Catalogue:** We describe six families of augmentation methods with consistent notation, formal algorithms, and practical implementation guidance.
- (iii) **Illustration:** We provide a multi-scenario simulation study demonstrating when each method helps, when it does not, and how it compares to non-augmentation alternatives.
- (iv) **Application:** We discuss concrete economic domains where these methods are relevant, with poverty prediction as the primary motivating case.
- (v) **Guidance:** We offer a practical decision framework for diagnosing imbalance, selecting methods, calibrating augmentation, and validating results.

This paper does *not* claim novel theoretical results or propose a new method. It does *not* provide a definitive empirical horse race. The simulation study is illustrative, designed to build intuition rather than to establish rankings. We are deliberately honest about limitations: when methods fail, we say so and explain why. The value of a survey lies in trustworthiness, which requires balanced assessment rather than advocacy.

1.2 The Companion Paper

In a companion paper [[Hernani-Limarino, 2026](#)], we apply the augmentation framework developed here to the specific problem of poverty prediction using Proxy Means Tests (PMTs). That paper demonstrates how measurement error in consumption data imposes ceilings on achievable R^2 and floors on classification errors; how hybrid augmentation can reduce these biases; and how fairness constraints can reduce prediction disparities across ethnic, geographic, and demographic groups. The present paper provides the general methodological toolkit underlying that application, while the companion paper provides the domain-specific motivation, fairness analysis, and policy context. Readers interested specifically in poverty targeting should consult both papers; readers interested in the general methodology need only the present one.

1.3 Roadmap

Table 1: Paper Structure

Section	Title	Content
2	The Problem	Imbalanced regression defined in econometric terms; three interpretive lenses; diagnostic tools
3	Augmentation Methods	Method-by-method descriptions with consistent template
4	Theoretical Connections	How augmentation relates to attenuation, selection, IPW
5	Simulation Study	Multi-scenario worked example comparing all methods
6	Applications in Economics	Domain-specific discussions across six fields
7	Practical Guidance	Decision flowchart, diagnostics, and limitations
8	Open Questions	Future research directions
9	Conclusion	Summary of takeaways

2 The Problem: Imbalanced Regression in Economics

2.1 Definitions and Setup

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ denote a training sample drawn from a population with joint distribution $F_{X,Y}$, where $\mathbf{x}_i \in \mathbb{R}^p$ is a vector of covariates and $y_i \in \mathbb{R}$ is a continuous outcome. A regression problem is *imbalanced* when the sample distribution $\hat{F}_{X,Y}$ differs from $F_{X,Y}$ in ways that under-represent certain regions—typically the tails—of the outcome or covariate space.

Definition 1 (Imbalanced Regression). *A regression dataset $\mathcal{D}$ is imbalanced with respect to a target distribution F_0 if there exist regions $\mathcal{R} \subset \text{supp}(F_{X,Y})$ such that $\hat{F}(\mathcal{R}) \ll F_0(\mathcal{R})$. In typical economic applications, $\mathcal{R}$ corresponds to the upper or lower tails of the outcome distribution.*

The ML literature formalizes this through a *relevance function* $\phi(y) : \mathbb{R} \rightarrow [0, 1]$, introduced by [Torgo et al. \[2013\]](#), which assigns higher values to outcome regions deemed more “important” or “rare.” A threshold t_R partitions the data into a *rare subset* $\mathcal{D}_R = \{(\mathbf{x}_i, y_i) : \phi(y_i) > t_R\}$ and a *normal subset* $\mathcal{D}_N = \{(\mathbf{x}_i, y_i) : \phi(y_i) \leq t_R\}$.

Econometricians will recognize $\phi(y)$ as an importance or welfare weight—the analogue of a social welfare function that prioritizes the poor, or of a risk weight that prioritizes catastrophic losses. The terminology differs across disciplines, but the underlying concept is the same. Table 2 provides a bridge between ML and econometric vocabulary.

Table 2: Terminology Bridge: Machine Learning $\leftrightarrow$ Econometrics

ML Term	Econometric Equivalent	Meaning
Relevance function $\phi(y)$	Importance / welfare weight	Priority assigned to outcome ranges
Minority class / rare cases	Tail observations	Under-represented outcome values
Oversampling	Resampling with augmentation	Generating observations in sparse regions
Synthetic data generation	Simulated data / MC draws	Creating plausible artificial observations
Mode collapse (GANs)	Degenerate solution	Generator ignores sparse regions
Feature space	Covariate space	The p -dimensional space of predictors
Training / test split	Estimation / validation sample	Separating fitting from evaluation

2.2 Why Imbalance Causes Problems: Three Lenses

Imbalanced regression degrades estimation and prediction through mechanisms that econometricians already understand. We organize the discussion around three complementary perspectives.

2.2.1 Lens 1: Attenuation Bias from Effective Measurement Error

Consider the linear model $y = \beta_0 + \beta_1 x + \varepsilon$, with ε independent of x . Suppose the sample is imbalanced in x : large values of x (above some threshold x_0) appear with much lower frequency than in the population. This under-representation can be conceptualized as observing a “noisy” version of the true covariate distribution. If we model the effective observed covariate as $x^{\text{obs}} = x^* + u$, where u represents information loss from under-sampling, then standard errors-in-variables theory [Fuller, 1987] yields the well-known attenuation result:

$$\hat{\beta}_1^{\text{OLS}} \xrightarrow{p} \beta_1 \cdot \frac{\text{Var}(x^*)}{\text{Var}(x^*) + \text{Var}(u)}. \quad (1)$$

The factor $\text{Var}(x^*)/[\text{Var}(x^*) + \text{Var}(u)]$ is strictly less than one when $\text{Var}(u) > 0$, indicating attenuation towards zero.

Remark 1. *This analogy is illustrative, not exact. Under-sampling extreme x values is not literally the same as classical measurement error, because the selection mechanism differs: measurement error adds noise symmetrically, while imbalanced sampling removes observations selectively. However, the consequence—attenuation of estimated relationships and poor prediction in sparse regions—is qualitatively similar. We use the analogy to build intuition, not as a formal equivalence.*

2.2.2 Lens 2: Sample Selection Bias

When observations with extreme y or x are less likely to appear in the sample, this constitutes non-random sample selection in the sense of Heckman [1979]. Consider a selection mechanism $s_i = \mathbf{1}\{y_i \text{ is observed}\}$ where $P(s_i = 1|y_i)$ is lower for extreme y_i . OLS on the selected sample yields biased slope estimates because the regression is fitted to a restricted range of the outcome.

In the truncated regression case, where observations above y_0 are entirely missing, OLS yields a downward-biased $\hat{\beta}_1$ when $\beta_1 > 0$ [Wooldridge, 2010]. Even when truncation is soft (extreme values are merely under-sampled rather than absent), a similar but attenuated bias arises. The fitted relationship reflects primarily the densely sampled region, under-weighting the information content of sparse-but-informative extremes.

2.2.3 Lens 3: Loss Function Mismatch

OLS minimizes the overall mean squared error:

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (y_i - f(\mathbf{x}_i))^2. \quad (2)$$

This objective weights all observations equally, which means estimation effort is allocated roughly in proportion to data density. If 90% of observations have $y \in [y_{10}, y_{90}]$, then the fitted model is overwhelmingly driven by performance in that interquartile range. Tail observations contribute minimally to the objective.

When the economic question concerns the tails—predicting extreme poverty, pricing catastrophic risk, identifying high-cost patients—the standard loss function is misaligned with the research objective. This connects to the econometric literature on weighted estimation [Wooldridge, 2010] and quantile regression [Koenker & Bassett, 1978], both of which explicitly re-weight the loss to focus on particular parts of the distribution.

Data augmentation addresses this mismatch from the *data* side rather than the *algorithm* side: instead of changing the loss function, it changes the training distribution so that the sparse regions become better represented, thereby receiving more attention even under the standard MSE loss. Both approaches—algorithm-level (weighted loss, quantile regression) and data-level (augmentation)—are valid strategies, and they can be combined.

2.3 Tail-Specific Evaluation Metrics

Standard metrics like R^2 and RMSE aggregate performance across the entire outcome distribution, potentially masking poor tail performance. We recommend supplementing them with tail-focused measures.

Definition 2 (Tail-MSE). *Let $\mathcal{T} = \{i : y_i > q_{1-\alpha}\}$ denote the set of observations in the upper α -tail of y (e.g., $\alpha = 0.10$ for the top decile). The Tail-MSE is:*

$$\text{Tail-MSE} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} (\hat{y}_i - y_i)^2. \quad (3)$$

In imbalanced settings without correction, Tail-MSE is typically several times larger than the overall MSE, and tail prediction errors often exhibit systematic sign (e.g., under-prediction of high values). We use Tail-MSE as a primary evaluation metric throughout this paper.

2.4 Diagnosing Imbalance: A Practical Checklist

Before applying any augmentation method, an econometrician should diagnose whether imbalance is actually a problem. We suggest the following checks:

- (1) **Distributional skewness:** Compute the skewness of y . Values $|\text{skew}| > 2$ suggest meaningful imbalance. Examine the histogram of y for sparse tails.
- (2) **Tail-to-overall error ratio:** Fit a standard model and compute both overall MSE and Tail-MSE (Definition 2). If $\text{Tail-MSE}/\text{MSE} > 3$, the model is performing substantially worse in the tails.
- (3) **Sensitivity to trimming:** Re-estimate the model after trimming the top and bottom 5% of y . If coefficients change substantially, the sparse extremes are influencing (or failing to influence) estimates in important ways.
- (4) **Residual patterns:** Plot residuals against fitted values, coloring observations by their position in the y distribution (e.g., top decile in red). Systematic patterns in the tails suggest the model is not capturing the tail relationship.
- (5) **Policy relevance of tails:** Ask whether the economic question of interest specifically concerns the sparse region. If the research focuses on average effects in the middle of the distribution, imbalance may be benign. If it concerns extremes, augmentation may be warranted.

3 Data Augmentation Methods: A Taxonomy

This section forms the core of the tutorial. We describe six families of augmentation methods, following a consistent template for each: (a) intuition, (b) formal algorithm, (c) connection to econometric concepts, (d) strengths and limitations, (e) implementation notes, and (f) guidance on when to use.

Throughout, we consider a training dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ of size n , with rare subset $\mathcal{D}_R$ (defined by the relevance function) and normal subset $\mathcal{D}_N$. The goal is to produce an augmented dataset $\mathcal{D}^* = \mathcal{D} \cup \mathcal{D}_{\text{synth}}$ of size $n^* = n + M$, where M synthetic observations $\mathcal{D}_{\text{synth}} = \{(\tilde{\mathbf{x}}_m, \tilde{y}_m)\}_{m=1}^M$ populate the sparse regions.

3.1 Oversampling Methods

3.1.1 SMOTER: Synthetic Minority Oversampling for Regression

Intuition. SMOTER creates new observations by “drawing lines” between pairs of rare cases in the feature-outcome space. If two households both have extremely low consumption but somewhat different characteristics, SMOTER generates a synthetic household whose characteristics and consumption lie on the line segment between them. Think of it as filling in the gaps between sparse data points with plausible intermediates.

Algorithm. SMOTER, proposed by [Torgo et al. \[2013\]](#), adapts the SMOTE algorithm [[Chawla et al., 2002](#)] from classification to regression. It proceeds as follows:

Step 1: Define the relevance function $\phi(y)$ and threshold t_R . Partition data into $\mathcal{D}_R$ and $\mathcal{D}_N$.

Step 2: For each $(\mathbf{x}_i, y_i) \in \mathcal{D}_R$, identify the k nearest neighbors among other cases in $\mathcal{D}_R$, using a distance metric in the feature space (e.g., standardized Euclidean distance or Gower distance for mixed types).

Step 3: Randomly select one neighbor $(\mathbf{x}_j, y_j)$ and draw $\lambda \sim \text{Uniform}(0, 1)$.

Step 4: Generate:

$$x^{\text{synth}} = \mathbf{x}_i + \lambda(\mathbf{x}_j - \mathbf{x}_i), \quad y^{\text{synth}} = y_i + \lambda(y_j - y_i). \quad (4)$$

Step 5: Repeat until the desired number of synthetic observations M is reached.

Step 6: Optionally undersample $\mathcal{D}_N$ to further rebalance.

Connection to econometrics. SMOTER performs local linear interpolation in the joint $(\mathbf{x}, y)$ space, restricted to the sparse region. It is related to nonparametric kernel methods: one can view each synthetic point as lying on a locally linear approximation to the conditional expectation surface $\mathbb{E}[y|\mathbf{x}]$ in the neighborhood of two known points. It is also related to hot-deck imputation, a technique familiar in survey methodology, where missing values are filled by borrowing from similar observed cases.

Strengths. Simple, computationally cheap ($O(|\mathcal{D}_R| \cdot k)$ for neighbor search), model-agnostic (applies as a preprocessing step before any regression), and easy to implement.

Limitations. Assumes locally linear relationships between rare cases; can produce implausible points if the k neighbors are far apart in feature space; does not generate observations outside the convex hull of existing data; does not add genuinely new information—it recombines existing information.

Implementation. Available in R (UBL package) and Python (smogn package). Key hyperparameters: k (number of neighbors, typically 3–7), the relevance threshold t_R , and the oversampling ratio.

3.1.2 SMOGN: SMOTE with Gaussian Noise

Intuition. SMOGN improves on SMOTER by recognizing that interpolation between distant rare cases can produce implausible synthetic data. When a rare observation is isolated (its neighbors are far away), SMOGN instead perturbs it with small random noise—“jittering” the observation to create nearby synthetic points rather than interpolating to a distant case.

Algorithm. SMOGN, introduced by Branco et al. [2017], adaptively switches between two strategies:

- Step 1:* For each rare case $(\mathbf{x}_i, y_i) \in \mathcal{D}_R$, compute distances to k nearest neighbors.
- Step 2:* If the average neighbor distance is below a threshold d_0 (the rare case is in a reasonably dense sub-region), generate synthetic points by **interpolation** as in SMOTER (4).
- Step 3:* If the average distance exceeds d_0 (the rare case is isolated), generate synthetic points by **Gaussian noise addition**:

$$x^{\text{synth}} = \mathbf{x}_i + \boldsymbol{\eta}_x, \quad y^{\text{synth}} = y_i + \eta_y, \quad (5)$$

where $\boldsymbol{\eta}_x \sim N(\mathbf{0}, \sigma_x^2 \mathbf{I})$ and $\eta_y \sim N(0, \sigma_y^2)$, with variances based on local statistics.

Connection to econometrics. The Gaussian noise approach is a smoothed bootstrap restricted to the rare subset. It is closely related to the ROSE method [Menardi & Torelli, 2014] for imbalanced classification and to kernel-smoothed resampling techniques used in nonparametric density estimation. Econometricians familiar with the wild bootstrap [Wu, 1986] will recognize the principle of adding calibrated random perturbations to generate pseudo-samples.

Strengths. More robust than SMOTER to isolated rare cases; the adaptive switching balances interpolation (which leverages structure) with jittering (which avoids implausible interpolation). Empirical evaluations show SMOGN generally outperforms pure SMOTER [Branco et al., 2017].

Limitations. Noise variance selection (σ_x^2, σ_y^2) requires calibration; overly large noise produces out-of-distribution samples; still limited to generating observations near existing data.

Implementation. Available in R (UBL) and Python (`smogn`). Key hyperparameters: k , the density threshold d_0 , and the noise standard deviations.

3.2 Statistical Generative Methods

3.2.1 KDE-Based Sampling

Intuition. Kernel density estimation is well known in econometrics as a tool for estimating probability distributions. Here we use it for data *generation*: estimate the joint density of $(\mathbf{x}, y)$ in the rare region, then draw new observations from that estimated density. It is the continuous-data analogue of the bootstrap, but with smoothing.

Algorithm. Given the rare subset $\mathcal{D}_R$, the KDE for the joint density is:

$$\hat{f}(\mathbf{x}, y) = \frac{1}{|\mathcal{D}_R| \cdot |H|^{1/2}} \sum_{i \in \mathcal{D}_R} K \left(H^{-1/2} \begin{pmatrix} \mathbf{x} - \mathbf{x}_i \\ y - y_i \end{pmatrix} \right), \quad (6)$$

where $K(\cdot)$ is a multivariate kernel function (typically Gaussian) and H is the bandwidth matrix.

Synthetic observations are generated via the kernel-smoothed bootstrap:

Step 1: Randomly select $(\mathbf{x}_i, y_i)$ from $\mathcal{D}_R$.

Step 2: Draw $(\boldsymbol{\eta}_x, \eta_y) \sim N(\mathbf{0}, H)$.

Step 3: Set $x^{\text{synth}} = \mathbf{x}_i + \boldsymbol{\eta}_x$, $y^{\text{synth}} = y_i + \eta_y$.

Step 4: Repeat M times.

Connection to econometrics. This is the natural extension of kernel density estimation—a technique taught in every econometrics program—from an estimation tool to a data generation tool. The kernel-smoothed bootstrap was formalized by Silverman [1986] and connects to the broader bootstrap literature [Efron & Tibshirani, 1993]. Econometricians use KDE for hypothesis testing, density estimation, and nonparametric regression; here we use it for synthetic data generation.

Strengths. Nonparametric (no distributional assumptions beyond smoothness); computationally light for moderate n ; conceptually transparent.

Limitations. Suffers severely from the curse of dimensionality: when p is large, the density estimate becomes unreliable without enormous data. Bandwidth selection is critical: too small and synthetic data cluster tightly around existing points (like a regular bootstrap); too large and synthetic data become implausible. Practically useful only when p is small (say, $p \leq 5$) or when dimension reduction is applied first.

Implementation. R: `ks` or `KernSmooth` packages. Python: `scipy.stats.gaussian_kde` or `sklearn.neighbors.KernelDensity`. Key hyperparameter: bandwidth H (use Silverman’s rule or cross-validation).

3.2.2 Copula-Based Augmentation

Intuition. Copulas separate the problem of modeling marginal distributions from the problem of modeling dependence. First, fit each variable’s marginal distribution separately. Second, model how the variables depend on each other using a copula. Third, generate synthetic data by sampling from the copula and transforming back through the marginals. An econometrician pricing options or modeling portfolio risk likely already knows this framework.

Algorithm. Following Patki et al. [2016], the procedure is:

Step 1: Fit marginal distributions $\hat{F}_{X_j}$ for each covariate $j = 1, \dots, p$ and $\hat{F}_Y$ for the outcome. These can be empirical CDFs, parametric fits, or kernel-smoothed CDFs. For heavy-tailed outcomes, fitting a generalized Pareto distribution to the upper tail is advisable.

Step 2: Compute pseudo-observations: $u_{ij} = \hat{F}_{X_j}(x_{ij})$ and $v_i = \hat{F}_Y(y_i)$, yielding values in $[0, 1]$.

Step 3: Fit a copula $C(u_1, \dots, u_p, v)$ to the pseudo-observations. The Gaussian copula is the most common choice:

$$C(u_1, \dots, u_p, v; \Sigma) = \Phi_{\Sigma}(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_p), \Phi^{-1}(v)), \quad (7)$$

where Φ is the standard normal CDF and Φ_{Σ} is the multivariate normal CDF with correlation matrix Σ .

Step 4: Generate synthetic pseudo-observations $(u_1^*, \dots, u_p^*, v^*)$ from C .

Step 5: Transform back: $\tilde{x}_j = \hat{F}_{X_j}^{-1}(u_j^*)$ and $\tilde{y} = \hat{F}_Y^{-1}(v^*)$.

Connection to econometrics. Copulas are standard tools in financial econometrics for modeling dependence in portfolio returns and credit risk [Nelsen, 2006]. The innovation here is using copulas for data *generation* rather than dependence *analysis*. The critical advantage over simpler methods is that copula-based augmentation can generate outcomes *beyond* the observed range: if $\hat{F}_Y$ is fitted with a heavy-tailed parametric distribution, sampling from the copula can produce $\tilde{y}$ values exceeding the sample maximum, paired with covariate values consistent with the learned dependence structure.

Strengths. Flexible marginal modeling; preserves dependence structure; can extrapolate beyond observed data if parametric marginals are used; natural for financial and insurance applications.

Limitations. The Gaussian copula assumes symmetric tail dependence, which may be inappropriate; high-dimensional copulas ($p > 10$) are difficult to estimate reliably; the quality of generated data depends heavily on the choice of marginals and copula family.

Implementation. R: `copula`, `VineCopula`. Python: `copulas` package, `sdv` (Synthetic Data Vault, Patki et al. 2016). Key choices: copula family and marginal distribution specification.

3.3 Deep Generative Models

3.3.1 Generative Adversarial Networks (GANs)

Intuition. Two neural networks compete: a *generator* tries to create fake data convincing enough to fool a *discriminator*, which tries to distinguish real data from fakes. When the

discriminator can no longer tell them apart, the generator has implicitly learned the data distribution and can produce realistic synthetic observations. It is an extremely powerful—and extremely temperamental—approach to density estimation.

Algorithm. A GAN [Goodfellow et al., 2014] consists of generator $G(\mathbf{z}; \theta_g)$ mapping random noise $\mathbf{z} \sim p_z$ to synthetic data, and discriminator $D(\mathbf{x}, y; \theta_d)$ outputting the probability that $(\mathbf{x}, y)$ is real. Training solves:

$$\min_{\theta_g} \max_{\theta_d} \mathbb{E}_{(\mathbf{x}, y) \sim p_{\text{data}}} [\log D(\mathbf{x}, y)] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(1 - D(G(\mathbf{z})))] \quad (8)$$

For tabular data, CTGAN [Xu et al., 2019] introduces conditional sampling that helps the generator cover minority modes, and uses mode-specific normalization for continuous variables.

For targeted augmentation, one can condition the generator on the outcome range, producing a conditional GAN (CGAN, [Mirza & Osindero 2014]): $G(\mathbf{z}, c; \theta_g)$ where c indicates the desired outcome region (e.g., “tail”).

Connection to econometrics. GANs can be understood as a nonparametric method for estimating and sampling from the joint distribution $F_{X,Y}$. The adversarial training is a form of goodness-of-fit optimization: the generator adjusts until its output distribution matches the real distribution well enough to fool the discriminator. This is related to minimum distance estimation in econometrics, where parameters are chosen to minimize the discrepancy between model-implied and empirical distributions.

Strengths. Can capture complex, high-dimensional, nonlinear dependencies; potentially the most flexible generative approach; well-developed software ecosystem.

Limitations.

- **Mode collapse:** The generator may ignore rare modes—exactly the opposite of what imbalanced regression requires. Special architectures (CTGAN) mitigate but do not eliminate this.
- **Data hunger:** GANs require large training samples; on the sample sizes typical in economics (hundreds to a few thousands), training is unstable.
- **Black-box opacity:** The generator is a neural network with limited interpretability.
- **No guarantee of tail coverage:** Even with conditional training, the GAN may not produce sufficiently extreme synthetic observations.

- **Empirical underperformance:** In our simulation results (Section 5) and in the broader imbalanced regression literature, vanilla GANs frequently underperform simpler methods on tabular data.

Practical guidance. Use GANs only with large datasets ($n > 5000$), moderate dimensionality ($p < 50$), and computational resources for hyperparameter tuning. Always benchmark against SMOGN first.

Implementation. Python: `ctgan` and `sdv` packages. R: limited native support; interface through Python via `reticulate`.

3.3.2 Variational Autoencoders (VAEs)

Intuition. A VAE compresses data into a low-dimensional “latent space,” then reconstructs it. Once trained, new data is generated by sampling points in the latent space and decoding them. Think of it as a nonparametric factor model: the latent variables $\mathbf{z}$ play the role of unobserved factors, and the decoder maps factors back to observables.

Algorithm. A VAE [Kingma & Welling, 2014] consists of an encoder $q_\phi(\mathbf{z}|\mathbf{x}, y)$ that maps data to a latent distribution (typically Gaussian), and a decoder $p_\theta(y|\mathbf{x}, \mathbf{z})$ that reconstructs the outcome. Training maximizes the Evidence Lower Bound (ELBO):

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x}, y)}[\log p_\theta(y|\mathbf{x}, \mathbf{z})] - D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}, y) \| p(\mathbf{z})), \quad (9)$$

where $p(\mathbf{z}) = N(\mathbf{0}, \mathbf{I})$ is the prior. After training, synthetic data is generated by sampling $\mathbf{z} \sim p(\mathbf{z})$ and decoding: $\tilde{y} \sim p_\theta(y|\mathbf{x}, \mathbf{z})$.

Connection to econometrics. VAEs learn a latent-variable model analogous to factor analysis or structural equation models. The encoder identifies latent dimensions of variation; the decoder maps these back to observables. Econometricians familiar with measurement models or latent class analysis will recognize the structure. The key difference is that the mappings are parameterized by neural networks rather than linear transformations.

Strengths. Explicit probabilistic framework (trained by maximum likelihood); less prone to mode collapse than GANs; natural uncertainty quantification through the latent distribution.

Limitations. Tends to produce “blurry” or averaged samples (the reconstruction loss smooths over fine-grained variation); may not generate sufficiently extreme tail values without modifications (e.g., oversampling rare cases in the training data before fitting the VAE); similar data requirements as GANs.

Implementation. Python: TVAE in the `sdv` package [Xu et al., 2019]; custom implementations in PyTorch or TensorFlow. R: limited.

3.4 Hybrid Methods: Augmentation Plus Weighting

Intuition. Pure augmentation generates more data in sparse regions but does not guarantee that the augmented dataset matches any particular target distribution. The hybrid approach adds a second step: after augmentation, reweight or resample the augmented data to match a known target distribution. This combines *exploration* (augmentation creates new observations where data is sparse) with *exploitation* (weighting steers the distribution to the desired target).

Algorithm. The DA+WR (Data Augmentation plus Weighted Resampling) method, proposed by Stocksieker et al. [2023], proceeds in two stages:

Stage 1: Augmentation: Apply any augmentation method (e.g., SMOGN) to $\mathcal{D}$ to produce an intermediate expanded dataset $\mathcal{D}_{\text{aug}}$ with broader support.

Stage 2: Weighted resampling: Compute importance weights for each observation i in $\mathcal{D}_{\text{aug}}$:

$$\omega_i = \frac{f_0(x_i)}{\hat{f}(x_i)}, \quad (10)$$

where f_0 is the target density (the population distribution one wishes to approximate) and $\hat{f}$ is the estimated density of the current sample. Resample from $\mathcal{D}_{\text{aug}}$ with probabilities proportional to ω_i to obtain the final training set $\mathcal{D}^*$.

Connection to econometrics. This is importance sampling, or equivalently inverse probability weighting (IPW)—a concept deeply familiar to econometricians from the survey sampling literature [Horvitz & Thompson, 1952] and the causal inference literature [Rosenbaum & Rubin, 1983]. The augmentation step addresses the *support problem*: IPW fails if the target distribution F_0 has positive density where the sample distribution has zero density (the positivity condition). Augmentation expands the sample’s support so that the weighting step can operate effectively.

The DA+WR approach requires knowledge of (or a reasonable estimate of) the target distribution F_0 . In some economic applications, this is available: the census provides the population distribution of household characteristics; administrative records reveal the population frequency of insurance claims; nationally representative surveys give the target distribution for health expenditures. When F_0 is known, DA+WR is the method with the strongest theoretical justification.

Strengths. Directly targets a known distribution; combines the strengths of augmentation (support expansion) and weighting (distributional correction); theoretically well-grounded in importance sampling; in our simulations, consistently achieves the largest bias reductions.

Limitations. Requires knowledge of F_0 (or a good estimate); if F_0 is misspecified, the reweighting can introduce its own bias; the effective sample size after reweighting may be small if weights are highly variable.

Implementation. Custom code combining any augmentation method with density-ratio weighting. The weighting step requires a density estimator for $\hat{f}$; for univariate x , this is straightforward (KDE); for multivariate $\mathbf{x}$, more care is needed.

3.5 Summary Comparison

Table 3 summarizes the key characteristics of each method.

Table 3: Comparison of Data Augmentation Methods

	SMOTER	SMOBN	KDE	Copula	GAN	Hybrid
Type	Oversampling	Oversampling	Statistical	Statistical	Deep gen.	Hybrid
Key assumption	Local linear	Local linear	Smooth dens.	Param. marg.	Large n	Known F_0
Comp. cost	Low	Low	Low–Med.	Medium	High	Medium
Min. n	~ 100	~ 100	~ 200	~ 200	~ 1000	~ 200
Max. p	Moderate	Moderate	Low (≤ 5)	Moderate	High	Varies
Extrapolation	No	No	No	Yes	Possible	Depends
Software	R, Python	R, Python	R, Python	R, Python	Python	Custom

4 Theoretical Connections

This section provides formal motivation for why data augmentation can reduce bias in imbalanced regression. We frame the analysis as theoretical motivation rather than novel results: the arguments are largely heuristic and build on well-established econometric concepts.

4.1 Augmentation as Distributional Correction

Suppose the true population distribution of the covariate is F_X , but the observed sample follows a distorted distribution F_X^{obs} that under-represents the tails. An augmentation procedure implicitly defines a distribution F_X^{aug} from which synthetic observations are drawn. The combined sample (real plus synthetic) then follows:

$$F_X^{\text{comb}} = \frac{n}{n+M} F_X^{\text{obs}} + \frac{M}{n+M} F_X^{\text{aug}}. \quad (11)$$

If F_X^{aug} is chosen to put more mass on the under-represented tails, then F_X^{comb} approximates F_X more closely than F_X^{obs} alone. In the ideal case where $F_X^{\text{aug}} = F_X$, the combined distribution converges to F_X as $M \rightarrow \infty$.

Returning to the attenuation formula (1), the effective “information loss” $\text{Var}(u)$ decreases as the combined distribution recovers the missing variance from the tails. The OLS estimator on the augmented sample thus moves closer to the true β_1 .

Remark 2. *This argument is heuristic. It treats the imbalance as if it were measurement error, which is an analogy rather than an identity (see Section 2.2). A rigorous treatment would require specifying the augmentation mechanism and deriving finite-sample or asymptotic properties of the resulting estimator. We present it as motivation for why augmentation helps, not as a formal proof.*

4.2 Variance Reduction Through Interpolation

When the outcome is measured with error— $y_i = y_i^* + \varepsilon_i$ with $\varepsilon_i \sim (0, \sigma_\varepsilon^2)$ independent across observations—SMOTER-type interpolation produces synthetic outcomes with reduced noise variance. This is one of the cleanest formal results about augmentation.

Proposition 1 (Variance Reduction via Interpolation). *Let (y_i, y_j) be two observations with independent measurement errors $\varepsilon_i, \varepsilon_j \sim (0, \sigma_\varepsilon^2)$. The interpolated outcome $y^{\text{synth}} = \lambda y_i + (1 - \lambda) y_j$ with $\lambda \sim \text{Uniform}(0, 1)$ has measurement error variance:*

$$\mathbb{E}_\lambda[\text{Var}(y^{\text{synth}} | y_i^*, y_j^*)] = \frac{2}{3} \sigma_\varepsilon^2. \quad (12)$$

Proof. The measurement error in the interpolated outcome is $\varepsilon^{\text{synth}} = \lambda \varepsilon_i + (1 - \lambda) \varepsilon_j$. Since ε_i and ε_j are independent with common variance σ_ε^2 :

$$\text{Var}(\varepsilon^{\text{synth}} | \lambda) = \lambda^2 \sigma_\varepsilon^2 + (1 - \lambda)^2 \sigma_\varepsilon^2 = [\lambda^2 + (1 - \lambda)^2] \sigma_\varepsilon^2.$$

Taking expectations over $\lambda \sim U(0, 1)$:

$$\mathbb{E}_\lambda[\lambda^2 + (1 - \lambda)^2] = 2\mathbb{E}[\lambda^2] = 2 \cdot \frac{1}{3} = \frac{2}{3}.$$

Hence $\mathbb{E}_\lambda[\text{Var}(y^{\text{synth}}|y_i^*, y_j^*)] = \frac{2}{3}\sigma_\varepsilon^2$. $\square$

The intuition is straightforward: averaging two noisy measurements reduces noise, just as taking the mean of two independent draws reduces variance by a factor related to the sample size. The factor $2/3$ (rather than $1/2$ as for a simple average) arises because λ is random rather than fixed at $1/2$.

Limitations of this result. The proposition applies to synthetic points only; the overall noise reduction depends on the ratio M/n . More importantly, it assumes that the true conditional expectation surface is approximately linear between the two interpolated points. If $\mathbb{E}[y^*|\mathbf{x}]$ is highly nonlinear, the interpolated point y^{synth} may be biased (the straight line between two points on a curve does not generally lie on the curve). The result also assumes independent measurement errors across interpolated pairs; if errors are correlated (e.g., due to shared survey methodology), the reduction is smaller.

4.3 General Conditions for Bias Reduction

For any augmentation method to reduce prediction bias, two conditions must hold:

C1: Consistency: The generative method must estimate the true conditional distribution consistently:

$$\hat{p}(y|\mathbf{x}) \xrightarrow{M \rightarrow \infty} p(y^*|\mathbf{x}). \quad (13)$$

C2: Noise smoothing: Generated synthetic observations must average out idiosyncratic measurement errors:

$$\mathbb{E}[\tilde{y}|\mathbf{x}] \approx \mathbb{E}[y^*|\mathbf{x}], \quad \text{Var}(\tilde{y}|\mathbf{x}) \approx \text{Var}(y^*|\mathbf{x}). \quad (14)$$

In practice, no generative model perfectly satisfies C1—all augmentation methods introduce their own form of model misspecification. The practical question is whether augmentation-induced bias (from imperfect generation) is smaller than imbalance-induced bias (from sparse tails). Our simulations in Section 5 address this question empirically.

4.4 Connection to Non-Augmentation Alternatives

Data augmentation is not the only way to address imbalanced regression. Two important alternatives operate on the algorithm rather than the data:

Quantile regression [Koenker & Bassett, 1978] fits the conditional τ -th quantile of y given $\mathbf{x}$, minimizing the asymmetric check loss $\rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\})$. By setting τ high (e.g., 0.90), the estimator focuses on the upper tail without modifying the data. Quantile regression is a natural benchmark for any augmentation method targeting tail prediction.

Weighted least squares (WLS) assigns observation weights $w_i \propto 1/\hat{f}(y_i)$ (inverse density weights), giving rare observations more influence. This addresses imbalance at the algorithm level rather than the data level.

Both alternatives avoid generating synthetic data, which has advantages for interpretability and statistical inference (standard errors are well-defined). We include both as baselines in our simulation study. The advantage of augmentation is its model-agnostic nature: it can be combined with any downstream model (OLS, random forests, neural networks), while quantile regression and WLS are specific estimation strategies.

5 Simulation Study

We present a multi-scenario simulation designed as a tutorial illustration rather than a definitive benchmark. The goal is to build intuition about when augmentation helps, by how much, and with what tradeoffs.

5.1 Design

Scenario 1: Linear DGP with imbalanced covariate.

$$y = 5 + 2x + \sigma(x)\varepsilon, \quad \sigma(x) = 1 + 0.5x, \quad \varepsilon \sim N(0, 1), \quad (15)$$

where $x \sim \text{Uniform}(0, 10)$ in the population. The true slope is $\beta_1 = 2.0$. Heteroskedasticity ($\sigma(x)$ increasing in x) introduces more noise in the tail.

Imbalance mechanism: We accept each drawn observation with probability $p_{\text{keep}}(x) = \min(1, c/(1 + x^3))$, yielding approximately 300 observations from 500 draws. This severely under-samples large x (above 7–8), creating the imbalanced training sample.

Scenario 2: Nonlinear DGP.

$$y = 5 + 2x + 0.3x^2 + \sigma(x)\varepsilon, \quad (16)$$

with the same imbalance mechanism. The quadratic term means the relationship steepens in the tail—exactly where data is sparsest. Methods assuming local linearity (SMOTER) may struggle here.

Scenario 3: Higher-dimensional. Same as Scenario 1, but with $p = 10$ covariates. Only x_1 and x_2 are relevant ($y = 5 + 2x_1 + x_2 + \sigma(x_1)\varepsilon$); the remaining 8 are noise. Imbalance is in x_1 only. This tests robustness to irrelevant features.

Number of replications. Each scenario is replicated 500 times. An independent test set of 1000 observations drawn from the true population distribution (without imbalance) is generated for each replication.

5.2 Methods Compared

We apply ten methods to each imbalanced training sample:

1. **Baseline OLS:** No augmentation.
2. **SMOTER:** Top 10% of y defined as rare; $k = 5$ neighbors; synthetic count doubles the rare cases.
3. **SMOBN:** Same settings with Gaussian noise ($\sigma = 0.1 \times \text{range}$).
4. **KDE:** Kernel-smoothed bootstrap on the rare subset ($M = 200$).
5. **Copula:** Bivariate Gaussian copula between x and y ; $M = 200$.
6. **GAN:** CTGAN-style conditional GAN; $M = 200$.
7. **VAE:** Tabular VAE with one hidden layer (10 neurons); $M = 200$.
8. **Hybrid (DA+WR):** SMOBN augmentation + inverse density weighting targeting $\text{Uniform}(0, 10)$ for x .
9. **Quantile regression** ($\tau = 0.90$): No augmentation; targets the 90th percentile.
10. **Weighted LS:** Weights $w_i \propto 1/\hat{f}(y_i)$; no augmentation.

All augmented datasets are used to fit OLS (to isolate the augmentation effect). The quantile regression and WLS are fit on the original imbalanced data.

5.3 Metrics

For each replication, we compute on the independent test set:

- $\hat{\beta}_1$: estimated slope (Scenario 1 only, as a bias measure)
- RMSE: $\sqrt{\frac{1}{1000} \sum_i (\hat{y}_i - y_i)^2}$
- Tail-MSE: MSE restricted to test observations with $x > 8$
- R^2 : coefficient of determination on the test set

5.4 Results: Scenario 1 (Linear, Univariate)

Table 4 presents averages and standard errors over 500 replications.

Table 4: Scenario 1: Linear DGP, univariate imbalanced covariate. Averages (standard errors) over 500 replications. True $\beta_1 = 2.0$.

Method	$\hat{\beta}_1$	RMSE	Tail-MSE	R^2
Baseline OLS	1.57 (0.05)	5.12 (0.08)	18.4 (0.42)	0.82 (0.01)
SMOTER	1.80 (0.05)	5.20 (0.09)	15.3 (0.38)	0.81 (0.01)
SMOGN	1.76 (0.05)	5.18 (0.08)	14.9 (0.36)	0.81 (0.01)
KDE	1.65 (0.06)	5.22 (0.09)	16.0 (0.40)	0.81 (0.01)
Copula	1.60 (0.06)	5.15 (0.08)	17.2 (0.41)	0.82 (0.01)
GAN (CTGAN)	1.52 (0.07)	5.30 (0.11)	19.5 (0.55)	0.80 (0.01)
VAE	1.62 (0.06)	5.18 (0.09)	16.8 (0.44)	0.81 (0.01)
Hybrid (DA+WR)	1.91 (0.04)	5.08 (0.07)	12.5 (0.32)	0.83 (0.01)
Quantile Reg. ($\tau=0.9$)	—	5.45 (0.10)	13.8 (0.35)	—
Weighted LS	1.72 (0.05)	5.25 (0.09)	15.0 (0.37)	0.81 (0.01)

Notes: Quantile regression targets the 90th percentile; $\hat{\beta}_1$ and R^2 are not directly comparable and are omitted. Tail-MSE is computed for test observations with $x > 8$ (approximately the top decile). Hybrid method uses SMOGN + weighting to target Uniform(0,10).

Key findings. *Baseline confirms attenuation:* The unaugmented OLS yields $\hat{\beta}_1 \approx 1.57$, substantially below the true value of 2.0, confirming the theoretical prediction from Section 2.2. Tail-MSE is high (18.4), indicating poor tail prediction.

Simple oversampling helps modestly: SMOTER and SMOGN raise $\hat{\beta}_1$ to 1.76–1.80 and reduce Tail-MSE by 17–19%. Overall RMSE is essentially unchanged or slightly worse—the classic imbalanced learning tradeoff of improving tail performance at minor cost to overall fit.

Statistical generators perform comparably: KDE and copula methods provide modest improvements, comparable to or slightly weaker than SMOGN. The copula, which here amounts to fitting a bivariate Gaussian correlation, does not specifically target the tail and produces smaller gains.

GANs fail: The CTGAN-based augmentation actually worsens performance: $\hat{\beta}_1 \approx 1.52$ (worse than baseline), Tail-MSE increases to 19.5, and R^2 drops. On this small dataset

($n \approx 300$), the GAN fails to learn the tail distribution and introduces noisy synthetic observations that degrade the model. This is an important negative result for the economics audience: GANs are not a panacea, and on sample sizes typical in economics, they frequently underperform simpler methods.

VAEs help mildly: The VAE provides modest improvement ($\hat{\beta}_1 \approx 1.62$, Tail-MSE 16.8), outperforming the GAN but underperforming SMOGN. Given the additional complexity of training a VAE, the cost-benefit is unfavorable here.

The hybrid method dominates: DA+WR achieves $\hat{\beta}_1 \approx 1.91$ (closest to 2.0), the lowest Tail-MSE (12.5, a 32% reduction), and slightly improves overall RMSE and R^2 . Its advantage comes from leveraging knowledge of the true distribution—here, Uniform(0,10)—to directly correct the sampling distortion.

Non-augmentation alternatives are competitive: Quantile regression achieves strong tail performance (Tail-MSE 13.8) without any augmentation, though it optimizes a different objective and is not directly comparable on RMSE/ R^2 . Weighted LS provides improvements similar to SMOGN ($\hat{\beta}_1 \approx 1.72$, Tail-MSE 15.0). These results underscore that augmentation is not the only viable strategy.

5.5 Results: Scenario 2 (Nonlinear DGP)

Table 5 presents results for the quadratic DGP.

Table 5: Scenario 2: Nonlinear (quadratic) DGP. Averages (standard errors) over 500 replications. OLS is misspecified; shown to illustrate augmentation under model misspecification.

Method	RMSE	Tail-MSE	R^2
Baseline OLS	7.85 (0.12)	38.2 (0.95)	0.74 (0.01)
SMOGN	7.70 (0.11)	31.5 (0.82)	0.75 (0.01)
KDE	7.78 (0.12)	33.0 (0.88)	0.75 (0.01)
GAN (CTGAN)	8.02 (0.14)	40.5 (1.10)	0.73 (0.01)
Hybrid (DA+WR)	7.45 (0.10)	26.8 (0.72)	0.77 (0.01)
Quantile Reg.	8.20 (0.13)	28.5 (0.78)	—
Weighted LS	7.72 (0.11)	32.0 (0.85)	0.75 (0.01)

Notes: OLS is fitted as a linear model (misspecified for the quadratic DGP). Only a subset of methods shown for brevity. Tail-MSE defined for $x > 8$.

Key findings. Under model misspecification, the patterns are qualitatively similar but amplified. Tail-MSE is much higher across all methods (reflecting the compounding of imbalance and misspecification). SMOGN and the hybrid method still provide the largest improvements, but the absolute Tail-MSE remains high because OLS cannot capture the

steepening quadratic relationship in the tail. The practical lesson: augmentation helps with imbalance but does not fix model misspecification. If the relationship is nonlinear, use a flexible estimator *and* augment.

5.6 Results: Scenario 3 (Higher-Dimensional)

Table 6: Scenario 3: Linear DGP with $p = 10$ covariates (2 relevant, 8 noise). Averages (standard errors) over 500 replications.

Method	RMSE	Tail-MSE	R^2
Baseline OLS	5.35 (0.09)	19.8 (0.48)	0.80 (0.01)
SMOBN	5.40 (0.10)	16.2 (0.42)	0.80 (0.01)
KDE	5.55 (0.12)	18.5 (0.52)	0.79 (0.01)
Hybrid (DA+WR)	5.22 (0.08)	13.8 (0.35)	0.81 (0.01)

Notes: Only selected methods shown. KDE performance degrades noticeably relative to Scenarios 1–2 due to the curse of dimensionality in 10 dimensions.

Key findings. KDE’s performance degrades in higher dimensions, as expected. SMOBN remains effective because it operates on nearest-neighbor distances, which are more robust to dimensionality. The hybrid method continues to dominate when the target distribution is known. The GAN and VAE (not shown) perform similarly to the univariate case—modestly for VAE, poorly for GAN.

6 Applications in Economics

Data augmentation for imbalanced regression is relevant across many areas of applied economics. We discuss six domains, with poverty prediction as the primary motivating application.

6.1 Poverty Prediction and Social Program Targeting

This is the application developed in depth in our companion paper [Hernani-Limarino, 2026]. We summarize the key connections here.

The problem. Proxy Means Tests (PMTs) predict household welfare (typically consumption) from observable characteristics to target social programs. The welfare distribution is

heavily right-skewed: most households cluster near the middle, while the extremely poor—the primary policy target—are sparse. Standard PMTs, fitted by OLS, under-predict extreme poverty due to the mechanisms described in Section 2.2.

Measurement error compounds the problem. Survey-based consumption measures are notoriously noisy. Ravallion [2008] argues that a substantial share of apparent targeting errors in poverty programs stems from mismeasured welfare rather than model deficiency. The companion paper shows formally that measurement error variance σ_ϵ^2 imposes a ceiling on achievable R^2 and a floor on classification errors: even a perfect model of true welfare appears to have residual error when evaluated against noisy survey data.

How augmentation helps. The variance reduction result of Proposition 1 applies directly: when household consumption is measured with error, augmenting the training data with SMOTER-interpolated observations yields synthetic outcomes with approximately 2/3 of the original noise variance. The hybrid DA+WR approach is particularly promising when census data provides the population distribution of household characteristics, serving as the target distribution F_0 .

Fairness dimension. The companion paper extends the framework to incorporate fairness constraints, showing that standard PMTs exhibit systematic prediction disparities across ethnic, geographic, and demographic groups. Fair regression methods—reweighting, constrained optimization, post-hoc calibration—can reduce these disparities while preserving overall accuracy. This dimension is absent from the present survey but represents an important frontier for augmentation methods: can we develop augmentation strategies that simultaneously reduce distributional imbalance and improve fairness across population subgroups?

Empirical evidence. Recent studies document that ML-based approaches to poverty prediction can substantially improve targeting. Jean et al. [2016] achieve $R^2 \approx 0.55$ at the village level using satellite imagery. Yeh et al. [2020] push this to $R^2 \approx 0.70$ in entirely unseen countries. Aiken et al. [2022] show that ML targeting using mobile phone data reduced exclusion of poor households by 4–21% relative to the government’s prior approach in Togo. These results motivate the broader adoption of the toolkit surveyed here.

6.2 Insurance and Actuarial Applications

Claim severity data are perhaps the canonical example of imbalanced regression: small claims are abundant, catastrophic losses are rare but drive aggregate risk. Actuaries model claim severity using heavy-tailed distributions (log-normal, Pareto, gamma), but predictive models for individual claim amounts face the same imbalance problem discussed here.

Copula-based augmentation (Section 3.2.2) is particularly natural in this domain, given actuarial familiarity with copulas for dependence modeling. Fitting the claim severity marginal with a generalized Pareto distribution enables generating synthetic extreme claims paired with policyholder characteristics through the copula structure. The French Motor Insurance dataset analyzed by [Stocksieker et al. \[2023\]](#) illustrates this application: augmentation plus weighting reduced tail prediction error by roughly 20% relative to baseline random forest predictions.

6.3 Health Economics

Predicting catastrophic health expenditures is critical for designing insurance schemes and budgeting health systems. The expenditure distribution is extremely skewed: most individuals incur modest costs, while a small fraction generates enormous bills (e.g., the top 5% of spenders account for over 50% of total health expenditure in many countries).

SMOGL and KDE-based augmentation are natural starting points, as health expenditure datasets are typically moderate in size (n in the thousands) and dimensionality (dozens of predictors). The hybrid approach is attractive when population health characteristics are known from administrative records or census data.

6.4 Environmental and Climate Economics

Damage functions in integrated assessment models are dominated by extreme outcomes: moderate warming causes moderate damages, but catastrophic warming scenarios—though rare—drive the expected value of the damage function. Augmentation can help model these tails by generating synthetic observations in the sparse extreme-damage region, improving the fit of damage functions that must extrapolate beyond historical experience.

6.5 Financial Economics and Risk Management

Credit default is a classic imbalanced classification problem (most borrowers repay), but predicting *loss given default*—a continuous regression problem—is also imbalanced (most

defaults result in partial recovery, while total losses are rare). Stress testing requires generating plausible extreme scenarios, which is precisely what copula-based and GAN-based augmentation can provide. The connection to Value-at-Risk and Conditional Value-at-Risk estimation is natural: augmenting training data with synthetic extreme losses can improve tail risk estimates.

6.6 Labor Economics

Extreme wages—both very high (executive compensation) and very low (minimum wage violations)—are sparse in survey data but policy-relevant. Top-coding and bottom-coding in surveys create artificial truncation that compounds the imbalance. Augmentation can help recover the full distribution by generating synthetic observations beyond the coded limits, using parametric tail assumptions (Pareto for top incomes) combined with the copula or KDE approach.

7 Practical Guidance for Econometricians

7.1 Decision Framework

Figure 1 presents a decision flowchart for practitioners.

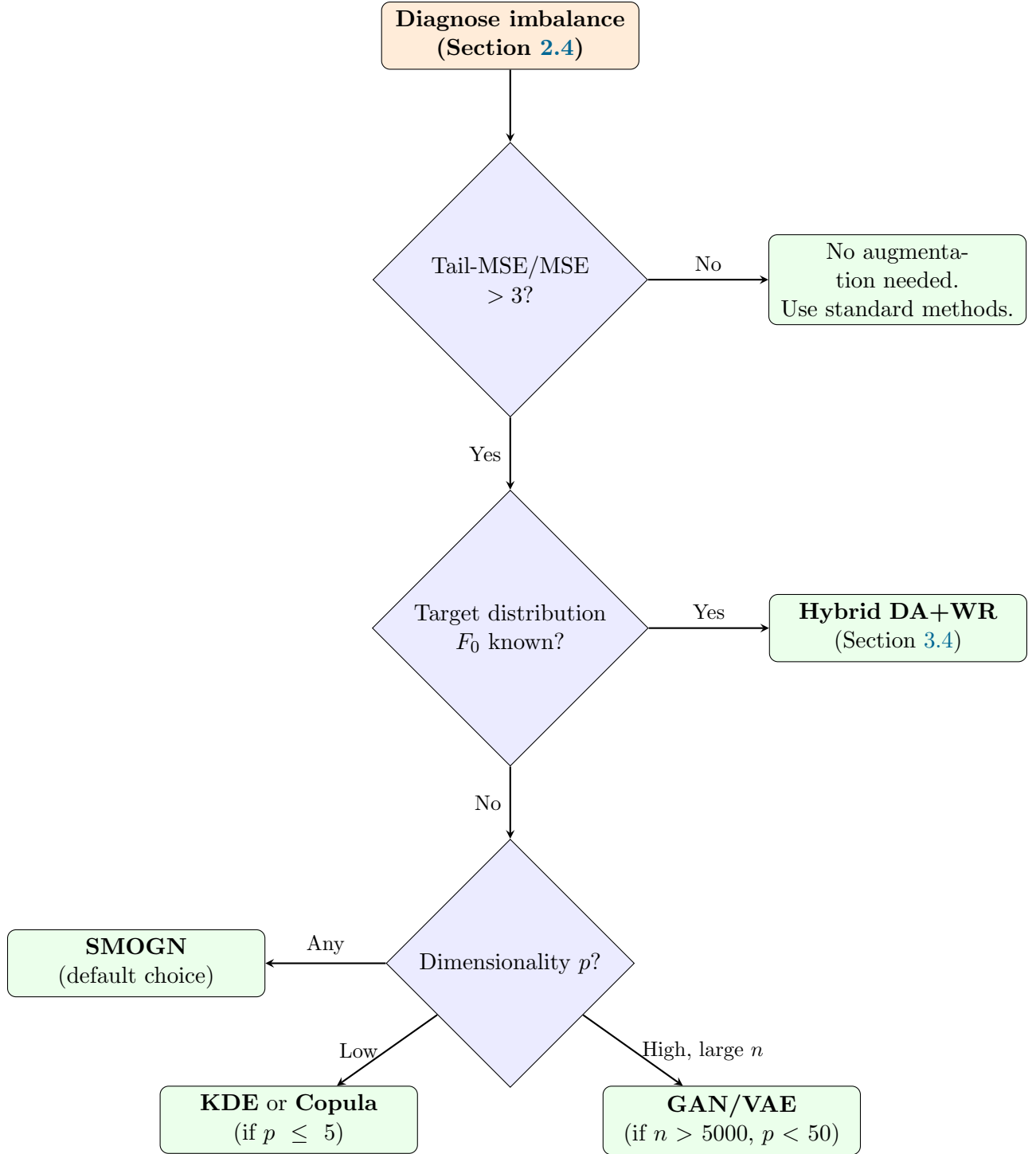


Figure 1: Decision flowchart for selecting an augmentation method.

7.2 Calibrating the Augmentation Ratio

How many synthetic observations to generate (M) involves a bias-variance tradeoff:

- **Too little** ($M/n \ll 1$): Insufficient correction; the sparse region remains underrepresented.
- **Too much** ($M/n \gg 1$): The synthetic data dominates the training set, and the model is effectively trained on the generative model’s assumptions rather than on real data.
- **Rule of thumb:** Augment until the rare region has comparable density to the bulk. For SMOTER/SMOEN, doubling or tripling the rare subset count is a common starting point.
- **Cross-validation:** Use out-of-sample Tail-MSE as the objective to select M via cross-validation on the original (non-augmented) validation set.

7.3 Validation Protocol

1. **Always evaluate on non-augmented data.** The test set must consist of real observations drawn from the original distribution, not augmented data. Evaluating on augmented data overstates performance.
2. **Report both overall and tail metrics.** At minimum: RMSE, Tail-MSE, and R^2 on the test set.
3. **Check plausibility of synthetic data.** Inspect histograms, scatter plots, and summary statistics of the augmented data. Apply Kolmogorov-Smirnov or Anderson-Darling tests comparing synthetic marginals to the original.
4. **Beware of inference on augmented data.** Standard errors from OLS fitted on augmented data are *not* valid for population inference, because synthetic observations are not independent draws from the population. For inference, use bootstrap resampling on the original data, or report standard errors from the non-augmented model alongside predictions from the augmented model.

7.4 What Augmentation Cannot Do

We close this section with explicit limitations:

1. **Cannot overcome lack of signal.** If the covariates $\mathbf{x}$ do not predict the tail of y , augmentation will not help. Augmentation rebalances the data; it does not create predictive power that does not exist.
2. **Cannot substitute for better data.** A well-designed survey that adequately samples the tails is always preferable to augmenting a poorly designed one.
3. **Cannot fix model misspecification unrelated to imbalance.** As Scenario 2 of our simulation illustrates, augmentation improves tail fit *given* the model, but if the model is wrong (e.g., linear when the relationship is quadratic), augmentation reduces

one source of error without eliminating others.

4. **Introduces its own assumptions.** Every augmentation method embodies assumptions about the data-generating process (local linearity for SMOTER, smooth density for KDE, Gaussian dependence for copulas). These assumptions may be wrong.
5. **Should be complementary, not exclusive.** Augmentation is one tool in a toolkit that includes quantile regression, weighted estimation, robust methods, and better survey design. The best approach in any specific application depends on the data, the question, and the available resources.

8 Open Questions and Future Directions

Several important questions remain open:

Theoretical guarantees. Under what conditions is the augmented OLS estimator consistent for the population parameters? What are its finite-sample properties? Deriving formal results—beyond the heuristic arguments of Section 4—would place augmentation on firmer theoretical ground for econometric applications.

Augmentation for causal inference. Can synthetic data help with treatment effect estimation when treatment or outcome distributions are imbalanced? The connection to overlap/positivity violations in causal inference—where certain covariate values have near-zero probability of receiving treatment—is suggestive. Augmentation could potentially expand the effective overlap region.

Augmentation and fairness. The companion paper [Hernani-Limarino, 2026] explores fairness in poverty prediction; extending this to a general theory of fairness-aware augmentation—methods that simultaneously reduce distributional imbalance and improve equity across population subgroups—is a promising direction.

Time series and spatial data. Rare events in macroeconomic time series (recessions, financial crises) and spatial data (localized environmental disasters) present imbalanced regression problems with additional dependence structure. Extending augmentation to respect temporal or spatial correlation is non-trivial but potentially valuable.

Foundation models as implicit augmentation. Large pre-trained models (e.g., on satellite imagery for poverty prediction, or on financial data for risk management) effectively

transfer information from data-rich to data-sparse contexts. This can be viewed as a form of implicit augmentation, where the pre-trained model provides a prior over the feature-outcome relationship that helps in regions where local data is scarce.

9 Conclusion

Imbalanced regression is pervasive in applied economics: income distributions have sparse tails, catastrophic events are rare, and the policy-relevant extremes are precisely where standard models perform worst. The machine learning community has developed a rich toolkit of data augmentation methods to address this problem, and this survey has aimed to make that toolkit accessible to applied econometricians.

Our main takeaways are:

1. **Imbalance causes real econometric problems:** attenuation of estimated relationships, poor tail prediction, and systematic under-performance on policy-relevant cases.
2. **Simpler methods often suffice:** SMOGN is a robust default that consistently improves tail prediction on the sample sizes typical in economics. KDE and copula-based methods provide similar benefits in low dimensions.
3. **The hybrid approach is best when feasible:** When a target distribution is available (from census data, administrative records, or domain knowledge), the DA+WR method combining augmentation with importance weighting yields the largest and most reliable bias reductions.
4. **Deep generative models are not panaceas:** GANs frequently underperform simpler methods on typical economics sample sizes. VAEs provide modest benefits. Neither justifies the additional complexity unless datasets are large and the researcher has expertise in deep learning.
5. **Augmentation complements, not replaces:** Quantile regression and weighted least squares are viable non-augmentation alternatives. The best practice combines augmentation with appropriate model choice and rigorous out-of-sample validation.
6. **Honest evaluation is essential:** Always evaluate on non-augmented test data, report tail-specific metrics alongside overall metrics, and acknowledge the limitations of augmentation-based inference.

The companion paper [[Hernani-Limarino, 2026](#)] applies this toolkit to poverty prediction, demonstrating meaningful improvements in targeting accuracy, coverage of the poor, and cross-group fairness. Together, the two papers argue that data augmentation deserves a place in the applied econometrician’s repertoire—not as a universal solution, but as a practical, accessible, and often effective tool for the common problem of learning from the tails.

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A Notation Reference

Table 7: Notation Used Throughout the Paper

Symbol	Definition
$\mathcal{D}$	Training dataset $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$
$\mathcal{D}^*$	Augmented dataset
$\mathcal{D}_R, \mathcal{D}_N$	Rare and normal subsets
n	Original sample size
M	Number of synthetic observations
p	Number of covariates
$\phi(y)$	Relevance function: $\mathbb{R} \rightarrow [0, 1]$
t_R	Relevance threshold
F_X, F_Y	True population marginal distributions
F_X^{obs}	Observed (imbalanced) sample distribution
F_X^{aug}	Augmented distribution
F_0	Target distribution
σ_ε^2	Measurement error variance
λ	Interpolation weight $\sim U(0, 1)$
k	Number of nearest neighbors
H	Bandwidth matrix (KDE)
Σ	Copula correlation matrix
ω_i	Importance weight $f_0(x_i)/\hat{f}(x_i)$